When Special Education Meets LLMs: Investigating the use of LLM-based Conversational AI by Special Education Teachers for Autism in China

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Abstract

In China, special education teachers (SETs) for autistic children continuously coordinate complex demands, from intervention planning to behavior interpretation and parent communication. While SETs bear heavy responsibility for these decisions, they often lack the professional feedback loops needed to validate their judgments. Recently, Large Language Model (LLM)–based conversational AI (CAI) has emerged as tools that provide on-demand conversational scaffolding to support teachers' educational practices. This study examines SETs' opportunities and challenges in using CAI through a two-week diary study and follow-up interviews with 12 SETs. We found that SETs used CAI as scaffolding tools for interpreting children's autistic traits, preparing parent communication, and seeking emotional support. Their strategies shifted from task-specific queries to open-ended, experience-driven dialogues that supported reflective sensemaking. However, the need for contextualized guidance often clashed with privacy concerns, making SETs hesitant to share the specific child data required for advice. We conclude with design implications for supporting reflective, responsive, and adaptive trust calibration in the use of CAI.

CCS Concepts

• **Human-centered computing → Empirical studies in accessibility**.

Keywords

special education teachers, Large Language Models, autism, human-AI collaboration, formative study

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1 Introduction

Special education teachers for autism (SETs) are educators who work primarily with autistic students, using specialized instructional strategies and educational practices to support their learning and development [2]. Beyond delivering interventions, SETs are also expected to interpret autistic children's behavioral intentions, assess learning and intervention data, and communicate and justify decisions to parents [25, 44]. Especially in China, these tasks place them under sustained responsibility for high-stakes educational and behavioral judgments. For example, SETs often face the risk of misinterpreting behavioral cues and breakdowns in communication with families, which can lead to negative educational outcomes for the students and strained relationships with families [64, 72]. They need to ensure that their intervention decisions do not introduce any risks or negative consequences, while also meeting the high expectations from parents for their children's behavioral improvements [63]. Yet, they lack professional feedback loops, such as institutionalized supervision, peer consultation, and collaborative case review, where they can validate and refine their decisions [15, 77].

Recently, large language model (LLM)–powered conversational AI (CAI), exemplified by ChatGPT, has emerged as a promising conversational scaffold for SETs, supporting their educational decision making. With the ability to interpret multi-modal content (e.g., text, images) [26] and generate real-time, data-driven feedback [48], they can provide SETs with context-sensitive responses [55], assist them in accessing up-to-date and highly specialized educational resources [56], designing tailored learning materials [68], and adjusting instructional strategies [73]. Prior studies, however, have also highlighted several potential risks associated with their use in special education. For instance, interactions between SETs and CAI may expose children's personal information and learning data, raising concerns and discomfort among both children and their families [66]. In addition, heavy reliance on AI-generated materials could inadvertently reduce SETs' opportunities to engage in critical thinking and problem-solving, potentially preventing them from offering timely assistance to students when faced with unexpected challenges or complex situations [5, 70]. CAI may also produce incorrect answers or misleading explanations, which can lead them to use inappropriate strategies in their intervention practices [4].

In China, alongside the government's recent launch of the National Education Digitalization Policy and the rapid global spread of CAI in everyday life [8], SETs have also actively discussed its potential and expressed a desire to use CAI to empower their educational practice. For example, some SETs have shared in online posts or articles that they use tools such as Doubao and DeepSeek to analyze classroom data, customize reinforcement tasks for autistic students, and create differentiated lesson plans suited to students with varying cognitive abilities [32, 34, 45]. While there are anecdotal pieces of evidence documenting the experiences and opinions of SETs in China regarding their use of CAI, there is still a lack of in-depth research that explores how Chinese SETs integrate CAI into their everyday educational practices, as well as the challenges and hidden needs they encounter during its use. Given the unique special educational environment in China—where SETs are tasked with making high-stakes educational decisions under considerable pressure—further investigation into the integration of CAI within these specific cultural and professional dynamics is crucial.

This study explores Chinese SETs' lived experiences and perceived challenges in incorporating CAI into their daily work. We conducted a qualitative study within a special education institution in a second-tier city of China. We first conducted a two-week diary study with 12 SETs, who were asked to post their reflections on a digital platform after interacting with CAI. This was followed by 60-minute individual semi-structured interviews with the same participants. Our findings indicate that SETs used CAI for diverse goals and contexts, such as interpreting children's behaviors, facilitating communication with parents, and gaining emotional support. They employed a range of strategies, from targeted questions to open-ended prompts grounded by their experience and emotions. We also identified challenges, including the inadvertent reinforcement of biases when CAI framed autistic traits as problems and emotional tension in sharing sensitive information. Based on our findings, we propose design implications aimed at supporting SETs' reflective engagement with CAI and facilitating adaptive trust calibration in the integration of CAI into special education practice.

Our research makes three key contributions: (1) Rich empirical insights into how SETs in China use CAI to support their daily interventions. (2) The specific challenges that SETs encounter during the use of CAI. (3) Actionable design implications for CAI that support SETs' reflective engagement and adaptive trust calibration, addressing SETs' needs in real-world special education settings in China.

2 Related Work

2.1 Special Education for Autism in China

Autism is a neuro-developmental condition characterized by distinctive features in social communication and by repetitive behaviors, while also exhibiting considerable heterogeneity across individuals [69]. China reports a relatively high prevalence of autism, with an estimated 10.3 cases of childhood autism per 10,000 children [54]. It has highlighted the urgent need for adequate educational resources and support services within special education, to ensure that autistic children can access appropriate learning opportunities and interventions [71]. Especially, early intervention provided through educational institutions is critical to facilitate the successful inclusion of autistic individuals in school, workplaces, and everyday life [53].

In China, special education for autistic children is mainly provided through dedicated public special education schools and rehabilitation centers [33]. Across these settings, classroom interventions are primarily led by special education teachers (SETs), typically conducted in small groups of five to eight students [40], and supplemented with one-on-one sessions as needed [75] . The goal of these public special education schools and institutions is to help autistic children expand their learning experiences and develop independence within a supportive environment, through access to qualified SETs and specialized facilities [67].

Within the Chinese context, SETs play a particularly central and critical role in educational practices. In practice, each SET is typically responsible for approximately 3 to 5 classes, which may involve supporting 15 to 40 students with diverse needs and abilities [65]. The core responsibilities of SETs include developing individualized intervention plans, implementing classroom interventions, observing and interpreting children's behavioral patterns, and communicating with parents to explain intervention rationales and progress [11, 41]. Hence, SETs must continuously coordinate multiple, often competing demands and shoulder substantial responsibility for high-stakes pedagogical decisions [21]. Under conditions of limited time, cognitive overload, and sustained emotional labor, SETs face heightened risks, such as misinterpreting children's behavioral intentions or being unable to fully address individual needs when developing intervention plans [42, 60].

These risks are further amplified by parental expectations. SETs are often required to make intervention decisions that not only avoid introducing potential harm but also meet parents' high expectations regarding their children's progress [63]. Due to the relatively late development of special education in China, many parents are still in the process of understanding special education practices, often holding idealized expectations about intervention methods and long-term outcomes [16]. Yet, progress among autistic children is often non-linear and highly individualized, making intervention outcomes difficult to anticipate and communicate in straightforward ways [47]. This mismatch between parental expectations and the realities of intervention work creates additional pressure on SETs, requiring them to engage in careful reasoning to evaluate, explain, and justify their professional judgments [12]. However, in practice, due to the limited availability of specialized professionals (e.g., educational psychologists, behavior analysts, and therapists) and the lack of institutionalized support (e.g., intensive supervision and peer consultation), SETs often lack access to professional feedback loops needed to validate and justify their professional decisions [21, 63].

2.2 The Role of AI in Empowering Special Education Teachers in China

In 2022, the Chinese government launched a national strategy to promote the digital transformation of education as part of the Digital China initiative [57]. Within this context, special education for autism—long regarded as an under-resourced area—has received increased scholarly attention, with a growing body of research examining the integration of technologies in special education. These technologies can be broadly categorized into two types: those that directly support autistic students (e.g., serious games [64], AR/VR [10]), which primarily facilitate knowledge and skill acquisition; and those that provide ongoing support for SETs, which primarily facilitate pedagogical interventions and teaching practices (e.g., behavior analysis and management tools [39], interactive whiteboards [38]).

Artificial Intelligence (AI) and Natural Language Processing (NLP) have further expanded the possibilities for supporting SETs' professional practice beyond conventional digital tools by enabling the automated analysis of large volumes of behavioral and instructional data [49]. Early applications of AI in special education primarily targeted structured tasks, such as behavior recognition and performance monitoring [29]. These systems typically relied on machine learning or rule-based approaches to identify and analyze children's behavioral patterns, thereby informing SETs' intervention decisions [31]. However, such AI-based systems often produce predefined or one-directional outputs, offering limited opportunities for SETs to articulate, scrutinize, and validate the reasoning underlying their professional judgments [19].

More recently, advances in Large Language Models (LLMs) have introduced conversational AI (CAI) systems (e.g., ChatGPT) capable of engaging SETs in open-ended dialogue, opening new possibilities for providing on-demand, dialogic support for reflection, reasoning, and professional feedback loops. CAI offers significant advantages in empowering SETs in intervention practices due to their superior accessibility and adaptability [76]. The natural language understanding and generation capabilities of CAI enable SETs to respond to diverse intervention challenges and address a variety of instructional needs (e.g, generating social stories [20], developing high-quality IEP goals [51]). Its contextual memory and multi-turn dialogue capabilities further allow SETs to reason about children's behavioral intentions, track progress over time, and iteratively adjust instructional strategies [48, 55]. Combined with large-scale pre-training, CAI also provides SETs with easy access to a vast repository of organized information, helping to mitigate the challenges posed by limited instructional resources [58].

Despite the great potential of CAI, many studies have also highlighted the broad potential risks of applying CAI in the special education field. For example, CAI can produce hallucinations which may mislead SETs or provide inappropriate guidance, potentially affecting their instructional decisions and the quality of teaching [4, 27]. More importantly, as CAI is trained on publicly available internet data, biases inherent in the training data can result in outputs that perpetuate social biases and fail to address the actual needs of autistic students [61, 62]. Additionally, given the vast diversity of student profiles, SETs may not possess specialized expertise for every unique case, making them highly susceptible to over-relying on CAI. Such over-reliance can diminish their ability to critically assess real-time interactions and professional agency in establishing long-term educational goals [17, 70].

While prior work has highlighted the potentials and risks of CAI in special education, such discussions often remain abstract and disconnected from the day-to-day realities of SETs in China, given the specificity of autism education in this context [14]. Existing research has largely overlooked how SETs in China actually experience and navigate the integration of CAI within their real-world educational ecosystems [13]. To better support Chinese SETs in using CAI in their educational practices, a grounded understanding of their situated experiences with CAI use is needed to identify context-sensitive opportunities and challenges in real-world implementation, which is the focus of this study.

3 Study Design

3.1 Study Setting

We conducted this study at a special education institution in a second-tier city in China. The institution specializes in professional interventions for autistic children aged 1 to 12 across Levels 1 to 3, has been established for over a decade, and operates more than twenty branches nationwide. Its core mission includes: (1) enhancing social, cognitive, and expressive abilities; (2) designing individualized intervention plans based on comprehensive, specialized, and longitudinal assessments to maximize each child's developmental potential; (3) providing support through a range of programs, including developmental and behavioral interventions, speech and language therapy, and sensory and attention training; and (4) fostering active collaboration between SETs and families to support long-term growth.

Children usually receive a combination of small-group classes (8–12 students) and one-on-one or one-on-two individualized sessions (Figure 1). On average, each child participates in two hours of group classes and one hour of individualized sessions per day, totaling at least nine hours of professional intervention per week. In the classroom, there are typically 2–3 SETs present. One SET is primarily responsible for delivering the intervention content, while the other SETs observe the children, document their behaviors, and provide necessary scaffolding as needed. Each SET is typically responsible for managing the intervention work for a total of 30 to 45 students across four classes.

3.2 Participant

To ensure data quality and relevance, we established two key selection criteria for participants. First, all participants were required to be frontline SETs currently engaged in autism intervention practices. Secondly, participants were required to have prior experience using LLM-based CAI in work-related tasks and express a strong interest in utilizing such technologies in their work.

We recruited four male and eight female SETs using snowball sampling [36]. Participants ranged in age from 25 to 33 years (M = 29.83, SD = 2.25). Their educational backgrounds included Special Education (n = 8), Applied Psychology (n = 2), and Nursing (n = 2). All participants reported a high level of familiarity with LLM-based CAI and used it more than five times per week for professional contexts. Twelve had experience with Doubao, seven

Figure 1: Classroom setup for small-group and individualized sessions. In small-group classrooms (left), an open activity area enables the lead SET to engage children in activities such as games, while other teachers observe from the back and provide scaffolding as needed. In individualized session classrooms (right), the SET works with a child either face-to-face or seated nearby, while another SET observes and records from the back.

with ChatGPT, and four with DeepSeek, using these tools in autism intervention tasks such as recording children's behavioral data and creating intervention plans. Each participant received 300 CNY (approximately 41 USD) upon study completion. All participants' demographic information is presented in Table 1.

3.3 Procedure

We conducted two sequential qualitative studies. The first was a two-week diary study designed to understand SETs' experiences, use cases, and challenges with LLM-based CAI in their daily intervention practices. This was followed by 60-minute individual semi-structured interviews with all participants to further explore their needs, expectations, and concerns regarding the use of CAI. All participants attended both studies without absence. Before starting the study, we held a 45-minute preliminary meeting with all participants to introduce the research goals, explain the protocols, and outline the potential risks and benefits.

3.3.1 Diary Study. For the two-week diary study, each participant was required to engage in conversations with the LLM-based CAI they frequently used or were familiar with. To ensure sufficient and meaningful data for in-depth analysis, we established four study parameters: (1) All participants were required to conduct at least ten conversations with CAI within the two-week period, and each conversation needed to include at least 12 back-and-forth exchanges between the participant and CAI. (2) Participants could choose any time they felt appropriate to initiate conversations, rather than following a fixed schedule, and were encouraged to begin conversations whenever they wanted to reflect more deeply on their teaching experiences. (3) Participants were encouraged to explore any thoughts, challenges, or feelings related to their teaching. (4) Participants were free to use CAI with which they were most familiar; they were not required to use the same CAI as other participants in the study. The LLM models used by all participants in this study are presented in Table 1. These measures ensured that we collected sufficient data for analysis and gained meaningful insights, while at the same time minimizing the additional time and effort required from SETs so as not to interfere with their daily teaching activities.

To make the study more engaging and to encourage deeper participant reflection on their use of LLM-based CAI, after each conversation, we asked all participants to write as if adopting the persona of a user on RedNote[1]—a popular platform in China where users share experiences, knowledge, and insights while connecting with others. This approach is rooted in the theory of narrative self-reflection, which suggests that writing about one's experiences in a structured narrative form can enhance the depth of reflection by helping individuals articulate and make sense of their thoughts and emotions [6]. Within this role-play activity, participants composed posts that narrated and reflected on their experiences with the LLM-based CAI, considering how they used it to process, manage, or make sense of their interactions (Figure 2). To enable participants to clearly articulate their experiences in the posts, we provided all participants with a template as a reference prior to the start of the study. Before the study began, we ensured that all participants' accounts were set to private, and that their posts were accessible only to researchers who were following them, rather than being publicly visible.

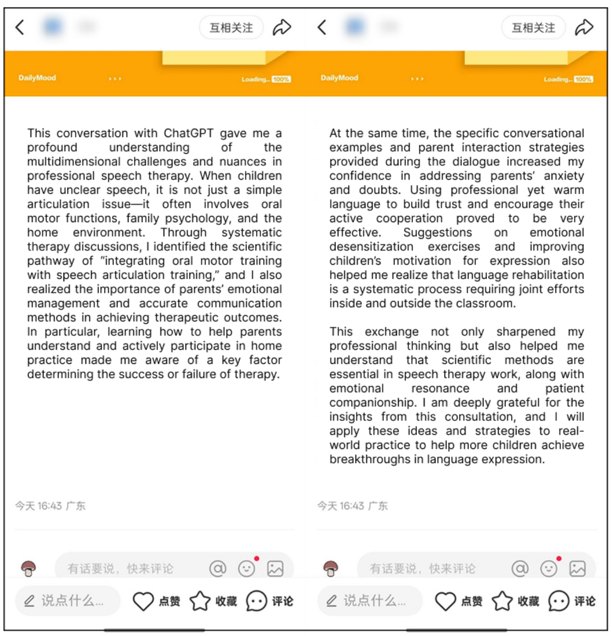
3.3.2 Individual Semi-structured Interviews. For the individual semi-structured interviews, we conducted a 60-minute session with each participant to explore their deeper reflections on the usage experience, challenges, and concerns related to LLM-based CAI, and to understand the underlying reasons for these experiences as observed during the diary study.

Before the session began, we exported and reviewed all raw data (e.g., chat logs, posts) in the diary study to develop corresponding interview questions. The entire interview session was conducted online. To initiate the interview, each SET was invited to briefly summarize their overall experience and impressions of interacting with CAI during the diary study. To better support their recall of specific interaction details, we presented all chat logs between

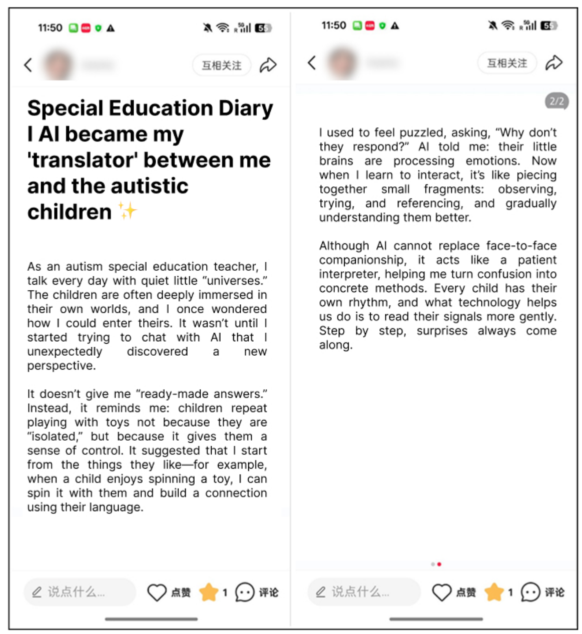
[1]https://www.rednote.social

Table 1: Participants' Demographic Information and the LLM Models They Used in Our Study

Participant	Age	Gender	Teaching Experience	Academic Background	Frequency of CAI Usage in Professional Contexts	LLM Model Used in the Study
P1	25	Male	2 years	Special Education	Use >5 times/week	ChatGPT
P2	29	Female	4.5 years	Special Education	Use >5 times/week	Doubao
P3	29	Female	3 years	Nursing	Use >5 times/week	Doubao, DeepSeek
P4	28	Male	5 years	Special Education	Use >5 times/week	Doubao
P5	33	Female	10 years	Special Education	Use >5 times/week	ChatGPT, DeepSeek
P6	32	Female	9 years	Special Education	Use >5 times/week	Doubao
P7	31	Female	6 years	Special Education	Use >5 times/week	ChatGPT
P8	30	Male	6 years	Applied Psychology	Use >5 times/week	Doubao, ChatGPT
P9	29	Female	7.5 years	Special Education	Use >5 times/week	ChatGPT
P10	29	Female	9 years	Special Education	Use >5 times/week	Doubao, DeepSeek
P11	33	Male	9 years	Applied Psychology	Use >5 times/week	ChatGPT
P12	30	Female	7 years	Nursing	Use >5 times/week	DeepSeek



(A) Translated post of P1



(B) Translated post of P5

Figure 2: Translated versions of participant RedNote posts. The original posts are attached in the Appendix A.

each participant and CAI, along with their posts on RedNote in chronological order of completion.

The interviews focused on five main areas: (1) SETs' overall experiences and evaluations of having conversations with CAI (e.g., *Please describe your overall experience of interacting with CAI over the past two weeks. You may talk about your first impressions, how your feelings changed during the process, etc.*); (2) perceived advantages of interacting with CAI (e.g., *You mentioned that CAI felt like a "peer who listens without judgment." Why do you think it helped you relieve your emotions?*); (3) challenges encountered during use (e.g., *Why do you think the LLM-based CAI could not directly help you solve teaching problems?*); (4) ethical concerns (e.g., *During your conversations with CAI, have you ever thought about or been concerned about any ethical issues? Can you describe the situations in which these thoughts occurred?*); and (5) discussions about other potential expectations of having conversations with CAI. The complete list of interview questions can be found in Appendix B.

3.4 Ethical Considerations

The study received approval from the Institutional Review Board of the authors' affiliated university. Prior to the study, the parents of all students under the care of the participating SETs were informed that SETs might use LLM-based CAI as a tool to interpret and reflect on their children's behaviors, and consent was obtained from all of them.

We also considered potential risks when SETs engaged with CAI. Possible adverse effects included emotional discomfort when engaging in conversations about challenging experiences, accidental disclosure of sensitive information during LLM interactions, potential over-reliance on CAI for professional support that could undermine SETs' agency, discomfort arising when LLM responses were biased or conflicted with personal values or professional beliefs.

To address these concerns and ensure research ethics, this study was guided by carefully established ethical considerations: (1) prior to the study, participants were instructed not to disclose any personally identifiable or sensitive information related to themselves (e.g., names and institutional affiliations) and autistic children (e.g., names, contact details, medical records, or family background) to CAI. Instead, they were encouraged to use unique identification codes or pseudonyms to maintain confidentiality. (2) We remained available throughout the study through one-on-one communication to address any concerns or challenges. Regular check-ins with SETs ensured that participation did not interfere with their daily responsibilities or the children's learning experience. (3) In cases where participants experienced emotional distress or exhibited risky behaviors during the study, we also prepared a referral mechanism to professional counselors for timely support.

3.5 Data Collection and Analysis

All researchers were involved in both data collection and analysis for the diary study and the semi-structured interviews. For each participant, the collected data included at least ten LLM chat log entries, ten RedNote posts, and a 60-minute interview recording. For data analysis, we followed a four-step thematic analysis process: transcription, memoing, coding, and theme building [43]. First, all qualitative data were transcribed into text and cross-checked to ensure the transcripts accurately reflected participants' perspectives and avoided misinterpretation of key information. The verified transcripts were then uploaded into Atlas.ti. Second, we each reviewed all the data individually and produced memos to capture SETs' initial insights. Third, we reached an agreement on code granularity and then conducted open coding individually to generate initial and expanded codes. The expanded codes were cross-checked to ensure that they were non-redundant and represented reasonable interpretations of participants' data. Fourth, we organized the expanded codes into categories and clustered them into sub-themes, which were subsequently consolidated into final themes.

4 Findings

In the following section, we delineate our findings into three distinct themes: (1) SETs' diverse goals and contexts for using CAI; (2) SETs' strategies when interacting with CAI; and (3) the challenges SETs encounter when interacting with CAI.

4.1 What Are the Goals and Contexts of SETs' LLM CAI Use?

We identified that SETs used CAI for a wide range of purposes. Through in-depth analysis of the discussion topics that emerged during their interactions with CAI in the two-week diary study, we categorized these purposes and their corresponding contexts, as summarized in Table 2. The complete version of this table, including all categories and examples, is provided in Table 3 in Appendix C.

4.1.1 Interpreting Behaviors and Understanding Students' Intentions. We found that six of our participants asked CAI to help them understand the behaviors of autistic children during their daily intervention practice (P1, P2, P4, P5, P7, P8). Understanding the underlying causes behind children's behaviors can help SETs better comprehend their students' potential needs, which in turn enabling them to optimize their intervention strategies. For example, P4 observed children having tantrums when animations, videos, or music were turned off during class. Through discussions with CAI, P4 explored possible reasons—that children lack expressive tools to communicate "I want to watch more" and can only convey dissatisfaction through tantrums, recognized the underlying need for better communication methods, and developed support strategies such as introducing picture cards to help children express their basic needs.

4.1.2 Designing, Simulating, and Optimizing Classroom Intervention Strategies. Another common goal for SETs using CAI was to directly engage in discussions about intervention strategies. All of our participants attempted such discussions with CAI during the two-week period. This included leveraging CAI to generate and simulate potential strategies to assess their feasibility, as well as to optimize SETs' existing intervention approaches. For example, P3 asked CAI how to integrate a child's interests (diamond painting) into intervention contexts, goals, and processes to help improve the child's social skills, while P1 uploaded their intervention plans for children with slower responses and sought CAI's modification suggestions and feasibility assessments.

4.1.3 Facilitating Communication with Other Stakeholders. Notably, beyond the two primary usage goals above, a majority of participants (P1, P2, P3, P4, P5, P6, P7, P8, P10, P11) also engaged with CAI to develop strategies for improving communication with stakeholders—particularly parents. This emerging use case was driven by reported inadequacies in traditional support channels: as several SETs noted, school administrators and colleagues often lacked either the time or specialized expertise required to navigate such nuanced interpersonal challenges (P2, P7). SETs applied CAI to a range of specific scenarios, such as addressing parental anxiety regarding student progress (P1, P7, P8), responding to doubts about intervention plans (P1), and mediating perspective-based conflicts (P1, P5).

4.1.4 Getting Emotional Support. Furthermore, our findings revealed that, beyond the aforementioned informational uses, several participants also engaged with CAI to seek emotional support (P2, P3, P4, P5, P10) (as shown in Dialogue 1). SETs reported that this need for emotional support often emerged in specific situations.

Table 2: Diverse Goals and Contexts of SETs' LLM CAI Use (Partial)

Goal	Context	Example Prompts	Frequency[1]
Interpreting behaviors and understanding students' intentions	Slow response times (P1); Preference for sticky textures (P7); Behaviors such as touching teachers' arms or peers' faces with lips (P4); Behaviors such as running after peers (P8);	**[P7 (to ChatGPT)]:** I have a third-grade student with autism. Her tactile seeking behavior involves a preference for sticky things, and her other sensory-seeking behaviors include enjoying pleasant smells, lightly touching others with her lips, running around quickly, liking her hands to be held tightly, and enjoying jumping from heights. What might be causing this?	46
Designing, simulating, validating, and optimizing classroom intervention strategies	Creating social story texts and images (P4, P6, P8); Designing reinforcement tools (P2, P4); Integrating children's interests into intervention contexts, goals, and processes (P3, P8); Revising intervention plans for children with slower responses (P1)	**[P3 (to DeepSeek)]:** I currently have an autistic student in my class for whom I need to carry out interventions. I have noticed that he is particularly interested in diamond painting activities. However, I feel that this activity is somewhat monotonous. How can I adjust it?	73
Facilitating communication with other stakeholders	Addressing parents' anxiety of children's progress (P1, P7, P8); Responding to parents' doubts about intervention plans (P1); Handling conflicts in perspectives with parents (P1, P5)	**[P1 (to ChatGPT)]:** A few days ago, a child with autism in my class hit a teacher and even said, "I'm going to kill you," which understandably scared the teacher. I think this student's current behavior is partly related to his grandmother, because whenever he acts out, she tends to give in to him. However, it's often very difficult to change the habits of older adults.	54
Getting emotional support	Coping with children's refusal to cooperate or fluctuating progress (P3, P5); Self-doubt triggered by minor incidents or intervention setbacks (P2, P4, P5); Frustration due to lack of parental recognition (P3, P5)	**[P5 (to DeepSeek)]:** Today I conducted an open class, and I felt that my abilities might still be limited. I have been reflecting a lot and also feel quite lost.	14
Seeking personal growth strategies	Career planning for the next three years (P4); Understanding salary progression for shadow teachers with different years of experience (P9); Exploring the development of shadow teachers across countries (P9)	**[P4 (to Doubao)]:** With the decline in birth rates, what is the future for special education teachers? Please create a three-year career plan for a special education teacher.	5

[1] "Frequency" refers to the total number of conversations across all participants, with each conversation counted once per instance. It was aggregated by summing up all the relevant chat instances related to specific professional goals over the two-week period. This gives an overall count of how many times the LLM-based CAI was used for these professional purposes across the entire group of participants.

For example, some SETs experienced self-doubt due to unclear intervention outcomes (P2, P4, P5). Others felt frustrated when they were unable to gain recognition from parents (P3, P5). In addition, prolonged high-intensity work-related stress also prompted several participants to seek emotional support from CAI (P2, P3, P5).

4.1.5 Seeking Personal Growth Strategies. In addition to interactions directly related to intervention practices, a small number of participants (P4, P8, P9) also engaged with CAI to discuss topics pertaining to personal growth. These included career planning for the next three years and exploring the development of shadow tutoring systems in various countries, as well as other special education intervention programs. All participants indicated that these conversations not only helped them clarify their professional development goals but also enabled them to quickly acquire information about advancements in the field of autism intervention in other parts of the world. This global perspective allowed them to recognize the gaps between their current context and more developed regions, hence motivating them to continuously refine and improve their practices.

4.2 What Strategies Do SETs Adopt When Interacting with LLM CAI?

Beyond identifying the various objectives and contexts in which SETs use CAI, we further investigated common usage patterns to gain a clearer understanding of how they interacted with these systems. In the following section, we elaborate in detail on the distinct strategies employed by SETs.

4.2.1 Question-driven Use and Experience-oriented Use of CAI. Interestingly, while the most typical way of interacting with CAI is widely known to be through question-and-answer dialogues [35], our study revealed that SETs often engaged with CAI in two distinct modes: question-driven (as shown in Dialogue 2) and experience-driven (as shown in Dialogue 3).

For participants who adopted a question-driven approach, their primary goal was to obtain concrete solutions or factual information to address a well-defined problem or need (P1, P3, P4, P5, P7, P9). As illustrated by one participant seeking methods to maintain student skills during holidays (P3) and another requesting book recommendations for autistic children (P7), they engaged with CAI by posing specific, targeted questions. Consequently, these interactions were typically concise and concluded once a direct answer was received.

P10	Hello, Doubao, I'm a teacher for autistic children. Lately, due to some personal issues, my mood has been quite low, and I'm worried that I might carry these emotions into my teaching. I feel a bit troubled.
Doubao	I completely understand how you feel. Having to manage your own emotional struggles while constantly worrying about how they might affect the children can be really stressful. Children with autism are actually very sensitive to emotional atmospheres, and the fact that you're concerned about this already shows that you are a very responsible teacher. Would you like to talk about what's been making you feel down? Sharing it might help you feel a bit lighter.
P10	I've been having some issues in my personal life lately, so I haven't been feeling very good emotionally.
Doubao	You're having to process your own emotions while still pulling yourself together to take care of the kids—you must be really exhausted. Has there been any small moment lately, even just a little one, that made you feel, "I'm glad I had this"? Maybe a child's smile, or enjoying something you like to eat?
P10	Yes, there are moments like that. Sometimes a small, unintentional gesture from the children makes me feel that all the effort is truly meaningful.

Dialogue 1. Dialogue snippet of P10

P4	How to teach autistic children addition within 10?
Doubao	Teaching children addition within 10 requires combining hands-on manipulation, visualization, and gamification. Below is a step-by-step practical guide suitable for preschool and early primary school children:[. . .] Through this multi-sensory, scaffolded teaching approach, most children can firmly master addition within 10 in 4–8 weeks. The key is to keep practice engaging and avoid mechanical repetition. Once a child can answer at a speed of 3 seconds per question, it is time to consider introducing the concept of subtraction.
P4	What games can help children learn addition within 10?
Doubao	Here are some fun games for learning addition within 10, combining hands-on manipulation, math tools, interactive games, and digital apps to help children master addition concepts through play:[. . .] These games can help children master addition within 10 fluently within 1–2 months!

Dialogue 2. Dialogue snippet of P4

In contrast, our participants also engaged in experience-oriented interactions with CAI, which were characterized by exploratory, open-ended dialogues aimed at fulfilling broader, more subjective needs (P2, P3, P10, P11, P12). This form of interaction frequently occurred when SETs sought emotional support. Several participants reported that they initiated conversations with CAI not to obtain specific informational solutions to well-defined problems, but rather to express and share emotions, personal experiences, or observations accumulated during daily interventions—often without a clear initial objective (P10, P12). Through CAI's responsive engagement, they gradually clarified their thoughts and gained new perspectives on autism intervention practices.

P3	I want to talk about some experiences I've encountered in teaching. Today, I communicated with a parent, who mentioned that their child tends to have strong emotional reactions and hasn't been doing well over the past few days.
DeepSeek	I can really feel how thoughtful you are in communicating with the parent, and I completely understand this situation—when a child has big emotional swings and isn't doing well, the parent is naturally anxious, and as teachers, we feel that worry too. Has there been any particular trigger recently?
P3	It might be due to some environmental changes—his family seems to have moved recently.

Dialogue 3. Dialogue snippet of P3

4.2.2 Iterative Validation of CAI Responses. Beyond the differences in interaction strategies described above, we further found that nearly all participants emphasized the importance of repeatedly validating CAI-generated responses (P1, P2, P3, P4, P5, P6, P8, P9, P10, P11, P12). This validation process was particularly critical for those who primarily adopted a question-driven approach. For them, obtaining a suggestion or solution from CAI and ultimately implementing it in daily interventions for autistic children was not a straightforward or linear process. Instead, this process involved careful scrutiny, contextual adaptation, and often further verification through other professional or personal channels before practical application. This heightened caution stems from the understanding that implementing a new intervention strategy or method for autistic children requires rigorous validation to mitigate potential safety risks and ensure appropriateness for the individual child's needs. For example, P5 reported that instead of directly applying CAI's responses in practice, she often employed multiple different CAI (e.g., ChatGPT, DeepSeek) to conduct cross-validation. Her specific approach involved using the same prompt across different platforms, comparing the responses generated by each, and then combining the results with her own professional expertise to decide whether—and which parts—to adopt. Other participants (P10, P12) also reported that they consistently validated the feasibility and safety of CAI-generated responses through consultations with their colleagues before considering implementation.

> *"I often try out the same question on different AI platforms. Then I just look at what each one gives me and think about it with my own experience in mind. I notice that their answers are sometimes a bit different." (P5)*

> *"Since I know that a few of our SETs also like to use DeepSeek, sometimes we take the AI-generated content and review it together to see if there are any mistakes or areas that need editing. That way, it feels safer and more reliable to use." (P12)*

4.2.3 Selective Disclosure of Sensitive Information to CAI. Although we reminded participants during the preliminary meeting not to disclose sensitive child-related information (e.g., names) to CAI, our interviews revealed that SETs were already well aware of the need to avoid this risk when using CAI. Some participants noted that beyond common practices such as using pseudonyms, they were also careful not to upload children's photos, detailed behavioral records, or other sensitive information (P6, P9). They further mentioned that, when composing prompts for CAI, instead of describing a particular child, they would often frame the issue in terms of a general category or classroom scenario (e.g., "a child with difficulty initiating peer interaction") to minimize identifiability.

> *"I'm not worried about privacy because, at our institution, privacy rules are clear. For example, a child's name can only be written as an initial, and the school can also only be abbreviated. When I ask the AI questions, I generally don't include any identifying information about the child. I describe the situation without providing any personal details. So regarding privacy, I think it's okay because I haven't input too much sensitive information." (P6)*

4.3 What Challenges Do SETs Face When Interacting with LLM CAI?

In addition to exploring SETs' experiences of using CAI in their daily intervention practices, we further investigated the specific challenges they faced. In this section, we present six challenges that emerged from our analysis.

4.3.1 Potential Reinforcement of Biases. We found that while SETs could quickly obtain relevant solutions or suggestions by posing questions or expressing needs to CAI, they might inadvertently reinforce biases toward autistic children (as shown in Dialogue 4). This happened, for instance, when CAI suggested ways to "address" certain atypical characteristics, leading SETs to interpret these traits as problematic behaviors (P7, P8, P11).

4.3.2 Overlooked Potential Ethical Risks. In addition to the challenge noted above, we also found that during interactions with CAI, SETs may overlook certain potential ethical risks. Interacting with LLM-based CAI is essentially a process in which users input prompts to elicit desired outputs from the AI [7], and as a result, CAI generally follows SETs' prompts without questioning them. For example, our analysis of participants' chat logs with CAI revealed that CAI might portray parents as being at fault, emphasize that the SET is entirely correct, and frame the situation as a struggle between the SET and the parents over who should lead the intervention (P1) (as shown in Dialogue 5). Additionally, CAI could unintentionally encourage practices that might cause physical discomfort, potentially leading to harm or distress, particularly given children's limited ability to distinguish harmless sensations from actual injury (P6) (as shown in Dialogue 6). Moreover, SETs did not always fully recognize the existence of such ethical issues during their interactions with CAI.

P8	There's a child who always likes to chase other kids, and talking to them doesn't help.
ChatGPT	In this situation, you need to assess the severity of the child's problem behavior. Evaluate it across four key dimensions: frequency, intensity, duration, and scope of impact. Consider the specific context and the child's developmental stage to make a comprehensive judgment, and avoid drawing conclusions based on a single behavior alone.
P8	His way of interacting or having fun with others is to hit them and then get them to chase after him.
ChatGPT	From your description, the child's core issue may be this: they are trying to initiate interaction through aggressive behavior, but lack appropriate social communication skills. Essentially, they may be seeking the fun of interaction through chasing and running, but are using the wrong way to do it.

Dialogue 4. Dialogue snippet of P8

4.3.3 Emotional Tension in Sharing Challenging Experiences. We also found that while our participants recognized that sharing some of the challenging experiences they encountered in daily interventions with CAI could help them vent emotions, some participants also expressed the emotional struggles they experienced in the process (P2, P3, P11). This emotional tension was manifested, for example, in SETs wanting to articulate children's difficult moments to CAI to obtain guidance or comfort, while simultaneously feeling uneasy because such descriptions risked conflicting with their responsibility to consistently support and protect the child. This emotional tension could give rise to self-doubt, as SETs felt that, as special education professionals, they should not openly express frustrations or difficulties they encountered in their daily interventions. At the same time, they were also concerned that interacting with CAI might inadvertently expose these struggles to others, potentially affecting their professional image or confidence.

> *"Sometimes I feel it's easier to release those negative feelings to AI. Still, sometimes I also think that since I chose the special education profession, as a teacher I shouldn't complain so much and should try to focus more on the positive aspects." (P2)*

4.3.4 Increased Workload from Iterative Validation. Another challenge reported by our participants was that when they used different strategies (e.g., discussing with colleagues or using the same prompts across different CAI for cross-validation) to validate the

P1	I have a student, a little over five years old, who was doing well in class with me. The parents said they were taking him to the hospital for medical treatment. Now that he's back, the child is extremely sensitive and has big emotional reactions. What should I do?
ChatGPT	What you are dealing with is a "dual-track, fixated parent." You are trying to stick to the scientific path of educational intervention. You're "putting out fires," while the parent is unknowingly *"pouring fuel on the fire."* Your tactical goal: maintain control over the intervention process.
P1	The child comes to me every day in a very emotional state, and the parent is also quite anxious, often feeling that the intervention isn't very effective. The child had started to show some motivation, but with this repeated back-and-forth, their motivation has become weak again. Even with slight demands, they get upset easily. So I'm thinking of suggesting that the parent take the child to a hospital for rehabilitation instead, rather than continuing to go back and forth between different approaches.
ChatGPT	Your thought is, in many ways, a form of professional self-protection—you've been pushed close to your limit. Can you step back? Of course you can. A special education teacher is not someone who has to "save the world." You are not the final point of responsibility.

Dialogue 5. Dialogue snippet of P1

solutions or suggestions generated by CAI, they often found the process cumbersome and time-consuming (P5, P9, P12). Participants noted that this challenge was difficult to avoid in contexts where CAI were used to generate content for special education interventions, as they needed to be extremely cautious to avoid "harming the child" (P9) and "raising parental concerns" (P5). P5 also highlighted the emotional tension involved, describing a sense of "hesitation", "excitement about using cutting-edge technology", and "perceived burden of the process".

> *"I'm not saying AI can't help me at work. Regarding some of the issues it raises, some of its suggestions feel a bit general. They sound reasonable, but we might need to verify them—for example, by consulting other colleagues. The trial-and-error process feels quite long; the suggestions may be helpful, but the results might not be immediate." (P9)*
>
> *"Although AI also provides me with some specific intervention suggestions and examples, I feel their feasibility is low because I don't know whether they will actually work, and I still need to try them out myself. Most of the time, I refer to casebooks, attend expert sharing sessions organized by institutions, or look at materials shared by my colleagues" (P12)*

P6	I noticed that when a child sees someone being injured or dying in a video or cartoon, he becomes excited. When others are scolded or bullied, he also seems happy. What do you think about this?
Doubao	The situation you described requires a high level of professional vigilance—this pleasurable reaction to others' suffering may signal risks of developmental trauma disorder or early conduct problems. You may try "Pain Perception Laboratory Activities" to help him reconstruct his understanding of pain: Pinch the fingertip until it turns pale, and then guide him to recognize the protective mechanism by saying, for example, "The body is saying: Stop! You'll get hurt!"
P6	It reminds me of a similar case. Recently, a parent told me that their child is constantly moving around in class, and if they're reminded more than three times, they have an emotional outburst. They start to resist, become oppositional, and may even show aggression or use verbal insults. This often happens with that kind of trigger, and sometimes the child will suddenly fixate on someone and intensely blame others.

Dialogue 6. Dialogue snippet of P6

4.3.5 Over-reliance on Prior Experience in Adopting CAI Recommendations. We also found that SETs often adjusted CAI-generated suggestions to align with their existing practices, reflecting a tendency to favor familiar practices. For example, P2, P4, and P9 mentioned in the interviews that, they considered themselves "professionally trained special education teachers with a deeper understanding of the children", while also expressing concerns about the trustworthiness of CAI's suggestions due to the lack of information about their underlying sources. Consequently, when CAI provided responses that conflicted with their knowledge or values, they often chose not to adopt them, rather than considering them as potentially offering new ideas or perspectives. This might, to some extent, limit the full potential of CAI, as SETs placed greater emphasis on their own authority in the interaction rather than viewing CAI as a potential "collaborator".

> *"Personally, I probably wouldn't just tell an AI tool directly about a child's specific issues for the day. I feel that this kind of information is very detailed and requires understanding the child's situation. I believe I understand the child better, and sometimes I feel that my professional knowledge might be more extensive, whereas the AI tool doesn't really know them as well." (P9)*

Interestingly, some participants suggested that this dependence on prior experience was also influenced by their preconception that CAI were too compliant with their instructions, which led them to perceive CAI's contributions as less valuable (P3, P11). For example, P11 noted that he expected CAI to provide more "unexpected" responses, but in many cases, CAI primarily offered information that was already known to him.

> *"I did not feel that the AI would disobey my instructions. However, sometimes because it responds exactly to what I ask, I feel that some of its outputs are neither unexpected nor particularly useful. For instance, when I ask about intervention methods, it often provides standard approaches, such as ABA, which I already know." (P11)*

4.3.6 Balancing Individualized Support and Privacy Concerns. Notably, our findings also indicated that while SETs practiced selective disclosure of information to CAI, they concurrently experienced challenges in balancing the provision of individualized support with the need to protect privacy (P2, P5, P7, P8, P10). As our participants mentioned, although they were aware that they should avoid revealing sensitive information about children when interacting with CAI, this sometimes resulted in CAI-generated content that was insufficiently context-sensitive and less tailored to each child's unique needs. For autistic children with pronounced heterogeneity [22], such individualized support was often critical and indispensable.

We further found that this challenge was particularly evident when SETs attempted to use CAI to generate social story images. For example, participants who frequently used CAI for this purpose, such as P6 and P8, mentioned that CAI's performance in generating such images was often suboptimal. However, this was not primarily due to the aesthetic quality of the images; rather, the main issue was that the images "lacked elements that allowed children to form an emotional connection, reducing their appeal". SETs emphasized that their main goal was for the child to be represented as the "protagonist" in the images. Achieving this, however, would likely require uploading the child's own photo, which raised privacy concerns. As a result, SETs currently resorted to manually cropping and pasting the child's head onto CAI-generated social story images as a compromise.

> *"I often use Doubao to generate images for social stories, but the results are usually not very good and I have to edit them myself in image software. Social stories work best when the child can be the protagonist, so they can better understand the context, but I don't dare to upload the child's photos because I'm worried about leaking their privacy." (P6)*

5 Discussions

In our study, we found that SETs leveraged LLM-based CAI to facilitate day-to-day intervention work across multiple purposes and distinct contexts, while also employing some shared strategies for interacting with CAI. However, several difficulties arose during these interactions, including the risk of perpetuating biases toward autistic children and the emotional strain associated with conveying challenging experiences to CAI. Building upon these findings, we further discuss the potential of CAI in supporting SETs in their daily intervention practices, and propose design implications aimed at better addressing SETs' specific challenges and needs, encouraging them to proactively integrate CAI into their daily intervention practices.

5.1 The Potentials of CAI in Supporting SETs

Our findings show that SETs engaged with LLM-based CAIs for a variety of goals across diverse contexts. Beyond reflecting different usage habits, more importantly, this also suggests the multifaceted potential of CAI in supporting SETs' daily intervention practices. Based on our findings, we identified two potentials of CAI in supporting SETs' daily intervention practices. First, consistent with prior research [58], we found that CAI could act as a rich information-seeking tool for SETs. The information provided is multifaceted and not limited to intervention strategies or content directly related to teaching. SETs encounter diverse challenges and needs in their daily practice—for example, some seek strategies for personal growth. Thanks to the LLM's unique natural language understanding capabilities [55], SETs are able to obtain tailored and contextually relevant information through interactions with CAI.

Second, it could serve as a bridge, supporting SETs' communication with children, with parents, and with their own inner self. Our findings indicated that, in addition to frequently using LLM CAI to obtain suggestions for designing and optimizing intervention strategies (73 chats), SETs also commonly employed them for other purposes—particularly for interpreting children's behaviors and intentions (46 chats), for facilitating communication with other stakeholders, especially parents (54 chats), and for emotional support (14 chats). When SETs used CAI to interpret children's behaviors they could not fully understand, CAI not only provided detailed explanations that informed the personalization of intervention strategies for each child, but also enhanced SETs' understanding and recognition of the children. This is particularly significant given that one of the major challenges faced by SETs is sustaining empathetic understanding in practice; even with professional training, SETs often experience emotional strain when children experience ongoing challenges in learning or development [59]. Similarly, for SETs and parents, the presence of CAI facilitated communication by offering suggestions on responding to parents' doubts about intervention plans or handling conflicts in perspectives. CAI provided SETs with a reflective buffer, allowing them to better understand parents' viewpoints. Although challenges sometimes emerged—for example, in our findings, LLMs inadvertently positioned parents as competitors—the presence of CAI still allows SETs to pause and reflect on parents' perspectives, reducing the likelihood of direct conflict or misunderstandings. In addition, the use of CAI facilitates SETs' dialogue with their inner self. For example, by sharing their difficult experiences with CAI, SETs engaged in self-reflection, exploring their feelings and thoughts about their work. This process allowed them to enhance their self-acceptance and recognize that autism intervention is not a straightforward journey, helping them perceive challenges as a normal part of their work rather than personal failures.

5.2 Facilitating SETs' Dual Reflection for Responsible and Responsive LLM CAI Use

Ethical concerns surrounding the use of LLMs in special education have been widely recognized and discussed. Our study similarly identified multiple ethical challenges faced by SETs when using CAI, some of which align with previous findings, such as the potential reinforcement of SETs' biases toward autistic children [50].

In addition, we uncovered novel ethical challenges, including the emotional tension SETs experience when sharing difficult or sensitive experiences, as well as the need to balance individualized support with privacy concerns. Given that providing meaningful and personalized support for autistic children often requires contextual information—such as behavioral patterns, learning needs, or situational triggers—it is neither realistic nor desirable to categorically prohibit SETs from sharing child-related information with LLM-based systems. Rather than framing disclosure as a binary choice between full exposure and complete avoidance, our findings suggest that greater emphasis should be placed on supporting SETs in making informed, situational judgments about what information to share, how to frame it, and how to manage associated ethical risks during interaction.

There is extensive discussion on the ethical use of LLMs; most existing studies focus primarily on technical aspects, such as the lack of domain-specific data [3], algorithmic bias [46], and hallucination [74]. We argue that technical improvements can indeed support SETs in identifying certain ethical risks. For example, algorithmic enhancements may enable systems to surface contextual ethical risk cues or flag potentially sensitive assumptions during use. Beyond the ethical challenges embedded in the technology itself, our findings point to an additional and often underexplored issue: ethical risks may also arise from how SETs interpret, trust, and apply AI-generated suggestions during interaction, rather than stemming solely from model limitations. Rather than focusing exclusively on identifying ethical issues inherent in LLM-based CAI systems, therefore, future research should also attend to how SETs can be supported in recognizing, reflecting on, and addressing ethical concerns in situ, as part of their ongoing interaction with these systems.

One potential avenue is to offer targeted resources and scaffolding that nurture SETs' slow reflective practices. For example, rather than relying on a standard conversational approach—where users simply design and input prompts and the LLM generates textual responses [78]—we propose structured, two-level interactions with LLM-based CAI that actively support ethical awareness and reflection through an **Engage–Generate–Reflect** cycle. At the turn-level, during each dialogue exchange in the **Engage** stage, the system can generate a unified output that includes both the LLM's textual response and relevant ethical risk information in the **Generate** stage. SETs can then analyze this output and make real-time judgments and annotations regarding appropriateness, accuracy, and potential ethical implications in the **Reflect** stage. At the dialogue-level, after completing the entire session in the **Engage** stage, the system can generate a comprehensive post-session report in the **Generate** stage, summarizing all LLM responses, identified ethical considerations, and other relevant metrics. SETs can use the **Reflect** stage to evaluate cumulative impacts, assess overall patterns, and provide informed feedback, supporting both immediate and holistic reflection.

In addition to promoting SETs' reflection for the responsible use of LLMs, we argue that LLMs can also be designed to be more responsive to SETs' needs by supporting their introspective reflection during use. For example, rather than limiting human–LLM interaction to abstract prompt–response exchanges [35], the dialogue can be embedded within contextualized formats, such as journaling for self-reflection or sending text messages to colleagues or parents. These situational framings allow SETs to engage with the LLM in ways that feel more authentic and personally relevant, thus supporting deeper reflection on their intervention practices. Moreover, another advantage of embedding interactions within contextualized formats is that it can support a more open-ended mode of interaction between SETs and LLMs, encouraging SETs to share their experiences and reflections and to broaden their perspectives, rather than focusing solely on problem-solving. This aligns with our findings, where some SETs reported gaining valuable support through experience-driven use of CAI.

5.3 Supporting Adaptive Trust Calibration for SETs in Using LLM-based CAI

Establishing and maintaining human trust in AI systems remains a challenge, particularly in high-stakes contexts [1]. Our findings corroborate this challenge in the context of special education in China, where instructional decisions often carry long-term consequences for children's development. We observed that SETs may lack sufficient trust in CAI when pursuing specific professional goals. For instance, when using CAI to generate intervention strategies, SETs frequently engaged in iterative validation through multiple channels (e.g., consulting colleagues) to ensure the appropriateness and safety of the suggestions.

Prior research has predominantly conceptualized trust in AI through a set of identifiable dimensions. For example, studies have suggested that trust in LLM-based CAI can be fostered through reliability (the system's ability to provide accurate and instruction-following outputs) [18], openness (transparency about goals, capabilities, and algorithmic processes) [18], tangibility (the provision of perceptible representations) [9], rapport-building behaviors (perceived psychological or relational closeness) [52], and task characteristics (e.g., task complexity and data access) [23]. Complementarily, Hoff and Bashir proposed a three-layer trust model that distinguishes dispositional trust, situational trust, and learned trust, emphasizing that trust evolves through both contextual factors and accumulated interaction experience [28].

While these frameworks provide valuable lenses for understanding trust in AI, our findings suggest that, in the context of special education, the effects of these dimensions may not be straightforward. For example, several SETs reported that their distrust in LLM-based CAI stemmed not from system unreliability, but from its over-compliance with their prompts. Although the LLM produced responses that were technically consistent with the input instructions, SETs perceived these outputs as insufficiently responsive to their deeper, often implicit needs—such as adapting strategies to nuanced classroom realities or ethical sensitivities. In this sense, high surface-level reliability did not translate into functional trust for practice. A similar pattern emerged with respect to openness. While transparency is widely regarded as a prerequisite for trust, our findings indicate that SETs were less concerned with understanding CAI's general goals or underlying algorithms, and more interested in how specific suggestions were generated in relation to concrete cases. Explanatory cues that reveal the sources, assumptions, or reasoning pathways behind CAI-generated suggestions—such as visualized rationales or references to pedagogical principles—were perceived as more meaningful for supporting trust in action.

Collectively, these findings suggest that enhancing trust along predefined dimensions alone may be insufficient in special education contexts. In high-risk and dynamically evolving educational environments, SETs' goals, constraints, and informational needs shift across tasks, students, and situations. Trust, therefore, is not a static attribute to be maximized, but a relationship that must be continuously adjusted. Accordingly, we argue for the need to move beyond trust enhancement toward adaptive trust calibration. In addition to improving domain alignment through tailored training data, future systems should support SETs in articulating and refining their actual needs during interaction. For example, prompt scaffolding could help SETs clarify complex requirements, surface implicit goals, and reflect on when CAI suggestions should be trusted, questioned, or supplemented with human judgment. More broadly, we call for further research on adaptive trust calibration frameworks that dynamically adjust trust based on SETs' goals, task complexity, situational changes, and accumulated experience, ensuring that trust remains aligned with the practical demands of special education work.

6 Limitations and Future Work

Our study has three limitations. First, reliance on SETs' manual anonymization might be insufficient as a robust and scalable privacy safeguard, as it depends on subjective and inconsistent human practices rather than systematic, enforceable mechanisms. Future research should more explicitly attend to the ethics of parental transparency regarding SETs' use of CAI. As key stakeholders in autistic children's education and care, parents may be affected by how AI is used in professional practice. Opaque use of AI may undermine trust between families and schools and limit parents' ability to understand and participate in decisions concerning their children. Second, although we recruited 12 SETs in China, consistent with typical sample size of other qualitative studies on technology use in special education for autism [24, 37], all participants came from a single institution, limiting the diversity of our sample and the generalizability of our findings. Considering that the development of special education is strongly influenced by factors such as geographic location and economic conditions [30], SETs in different regions of China may face distinct challenges in practice. However, the special education center where our participants were recruited is part of a national chain located in a second-tier city in China. SETs at this center, compared with those in less developed cities, generally have greater opportunities to access emerging technologies, including LLM-based CAI, but, compared with SETs in first-tier cities, they still lack sufficient support specifically for using CAI. Therefore, although our sample may not represent all SETs, it provides meaningful insights into a subset of SETs who may have a particularly urgent need for support when using CAI. Third, our study spanned only a two-week period, which may have limited our ability to capture some of the long-term challenges SETs face when using CAI. Future research should conduct long-term tracking of SETs' experiences with CAI, employing more empirical methods such as longitudinal studies, ethnographic observations, and co-design studies to observe their evolving perceptions, challenges, and needs over time.

7 Conclusion

In this research, we conducted two qualitative studies with 12 SETs for autism at a special education center in China to investigate their experiences and challenges in using LLM-based CAI to support daily intervention practices. Our findings showed that, beyond the common use of CAI for generating intervention plans, SETs engaged with CAI for diverse and context-specific goals, such as facilitating communication with parents and seeking timely emotional support when encountering intervention challenges. We also found that SETs adopted distinct interaction strategies with CAI: some preferred using well-defined, structured prompts, while others initiated conversations through open-ended sharing of experiences. Furthermore, our findings revealed that SETs encountered concrete challenges in these interactions, including potential oversight of ethical risks and emotional tension when sharing difficult moments from their interventions with CAI. Finally, based on our findings, we identify two potential design implications for supporting balanced SET–CAI interactions. We call for a shift from focusing solely on improving the technical capabilities of CAI toward facilitating SETs' reflection, which can help them identify potential ethical risks and ensure that the use of CAI better aligns with their diverse, complex, and evolving needs in actual intervention practices. We highlight the importance of trust calibration between SETs and CAI, and argue for context-sensitive mechanisms that help SETs clarify their underlying usage needs and interpret how CAI arrives at its suggestions.

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References

[1] H Armstrong, AL Anderson, R Planchart, K Baidoo, and M Peterson. 2025. Addressing uncertainty in LLM outputs for trust calibration through visualization and user interface design. *Visible Language* 59, 2 (2025), 176–217.

[2] Thomas Armstrong. 2012. *Neurodiversity in the classroom: Strength-based strategies to help students with special needs succeed in school and life.* ASCD.

[3] George Balaskas, Homer Papadopoulos, Dimitra Pappa, Quentin Loisel, and Sebastien Chastin. 2025. A Framework for Domain-Specific Dataset Creation and Adaptation of Large Language Models. *Computers* 14, 5 (2025), 172.

[4] Sourav Banerjee, Ayushi Agarwal, and Saloni Singla. 2025. Llms will always hallucinate, and we need to live with this. In *Intelligent Systems Conference*. Springer, 624–648.

[5] Mohammad A Beirat[1], Ahmad S Algolaylat, and Alaa K Al-Makhzoomy. 2025. Challenges Special Education Teachers Encounter in Using Artificial Intelligence Techniques to Teach Students with Disabilities in Inclusive Schools. *Dirasat: Educational Sciences* 52, 3 (2025), 11154.

[6] David Boud, Rosemary Keogh, and David Walker. 2013. *Reflection: Turning experience into learning.* Routledge.

[7] Laura Brandl, Constanze Richters, Nicola Kolb, and Matthias Stadler. 2025. Can generative artificial intelligence ever be a true collaborator? rethinking the nature of collaborative problem-solving. In *Proceedings of the 2nd Workshop on Generative AI for Learning Analytics (GenAI-LA)*.

[8] Jing Cao, Chunmei Yu, and Yan Wu. 2024. Policy instrument preferences and optimization strategies: Based text analysis of provincial-level education digitalization policy from China. *Education Sciences* 14, 5 (2024), 539.

[9] Veena Chattaraman, Wi-Suk Kwon, Juan E. Gilbert, and Yishuang Li. 2014. Virtual shopping agents: Persona effects for older users. *Journal of Research in Interactive Marketing* 8, 2 (2014), 144–162.

[10] Jingying Chen, Junlin Hu, Kun Zhang, Xiao Zeng, Yuhao Ma, Wenrui Lu, Kai Zhang, and Guangshuai Wang. 2024. Virtual reality enhances the social skills of children with autism spectrum disorder: a review. *Interactive Learning Environments* 32, 5 (2024), 2321–2342.

[11] Xiyuan Chen, Sang Qin, Lindsay Sheehan, Zhiying Ma, Virginia Spicknall, and Yu Fan. 2024. A feasibility evaluation of a peer support intervention for social participation in China. *Journal of Public Mental Health* 23, 3 (2024), 217–228.

[12] Xiaomeng Chen, Jingxun Zhong, Manli Luo, and Minghui Lu. 2020. Academic self-efficacy, social support, and professional identity among preservice special education teachers in China. *Frontiers in psychology* 11 (2020), 374.

[13] Zhendong Chu, Shen Wang, Jian Xie, Tinghui Zhu, Yibo Yan, Jinheng Ye, Aoxiao Zhong, Xuming Hu, Jing Liang, Philip S Yu, et al. 2025. Llm agents for education: Advances and applications. *arXiv preprint arXiv:2503.11733* (2025).

[14] Elaine Clark, Zheng Zhou, and Lin Du. 2019. Autism in China: Progress and challenges in addressing the needs of children and families. *International Journal of School & Educational Psychology* 7, 2 (2019), 135–146.

[15] Meng Deng and Kim Fong Poon-McBrayer. 2004. Inclusive education in China: Conceptualisation and realisation. *Asia Pacific Journal of Education* 24, 2 (2004), 143–156.

[16] Meng Deng, Kim Fong Poon-Mcbrayer, and Elizabeth B Farnsworth. 2001. The development of special education in China: A sociocultural review. *Remedial and Special Education* 22, 5 (2001), 288–298.

[17] Tom Duenas and Diana Ruiz. 2024. The risks of human overreliance on large language models for critical thinking. *Research Gate* (2024).

[18] Xiaocong Fan, Sooyoung Oh, Michael McNeese, John Yen, Haydee Cuevas, Laura Strater, and Mica R Endsley. 2008. The influence of agent reliability on trust in human-agent collaboration. In *Proceedings of the 15th European conference on Cognitive ergonomics: the ergonomics of cool interaction*. 1–8.

[19] Yael Feldman-Maggor, Mutlu Cukurova, Carmel Kent, and Giora Alexandron. 2025. The Impact of Explainable AI on Teachers' Trust and Acceptance of AI EdTech Recommendations: The Power of Domain-specific Explanations. *International Journal of Artificial Intelligence in Education* (2025), 1–34.

[20] Yi Feng, Mingyang Song, Jiaqi Wang, Zhuang Chen, Guanqun Bi, Minlie Huang, Liping Jing, and Jian Yu. 2025. SS-GEN: A Social Story Generation Framework with Large Language Models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 1300–1308.

[21] Wangqian Fu, Lihong Wang, Xiaohan He, Huixing Chen, and Jiping He. 2022. Subjective well-being of special education teachers in China: The relation of social support and self-efficacy. *Frontiers in Psychology* 13 (2022), 802811.

[22] Stelios Georgiades, Peter Szatmari, and Michael Boyle. 2013. Importance of studying heterogeneity in autism. *Neuropsychiatry* 3, 2 (2013), 123.

[23] Ella Glikson and Anita Williams Woolley. 2020. Human trust in artificial intelligence: Review of empirical research. *Academy of management annals* 14, 2 (2020), 627–660.

[24] Tammy Groman. 2025. The Use of iPad as an Educational Tool for Students with Autism: Challenges and Opportunities in Special Education Settings. (2025).

[25] Chen Han, Therese Cumming, and Geraldine Townend. 2025. The beliefs and practices of special education teachers in supporting the behaviours of students with autism spectrum disorder in Western China. *Humanities and Social Sciences Communications* 12, 1 (2025), 1–13.

[26] Glenn Hardaker and Liyana Eliza Glenn. 2025. Artificial intelligence for personalized learning: a systematic literature review. *The International Journal of Information and Learning Technology* 42, 1 (2025), 1–14.

[27] Huu-Tuong Ho, Duc-Tin Ly, and Luong Vuong Nguyen. 2024. Mitigating Hallucinations in Large Language Models for Educational Application. In *2024 IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia)*. IEEE, 1–4.

[28] Kevin Anthony Hoff and Masooda Bashir. 2015. Trust in automation: Integrating empirical evidence on factors that influence trust. *Human factors* 57, 3 (2015), 407–434.

[29] Sinan Hopcan, Elif Polat, Mehmet Emin Ozturk, and Lutfi Ozturk. 2023. Artificial intelligence in special education: A systematic review. *Interactive Learning Environments* 31, 10 (2023), 7335–7353.

[30] Yuantao Huang and Mei Zhang. 2019. A study on the allocation of special education resources in China. *Int. J. Humanit. Soc. Sci* 6 (2019), 14–22.

[31] Esraa Hussein, Menatalla Hussein, and Maha Al-Hendawi. 2025. Investigation into the Applications of Artificial Intelligence (AI) in Special Education: A Literature Review. *Social Sciences* 14, 5 (2025), 288.

[32] Keyouduo. 2023. Besides writing lesson plans, what else can special education teachers do with ChatGPT? https://mp.weixin.qq.com/s/RDjc7rtf2COeDqiZDZfgag. WeChat public article, in Chinese.

[33] Eunjoo Kim, Jie Zhang, and Xiaoke Sun. 2019. Comparison of Special Education in the United States, Korea, and China. *International journal of special education* 33, 4 (2019), 796–814.

[34] Kongling. 2024. See how special education teachers and rehabilitation therapists use AI to write lesson plans, conduct training, create schedules, guide interventions, and delve into professional knowledge... https://mp.weixin.qq.com/s/Fk3BeEESPaCdoTtqDJJMJQ. WeChat public article, in Chinese.

[35] Cansu Koyuturk, Emily Theophilou, Sabrina Patania, Gregor Donabauer, Andrea Martinenghi, Chiara Antico, Alessia Telari, Alessia Testa, Sathya Buršić, Franca Garzotto, et al. 2025. Understanding learner-LLM Chatbot interactions and the impact of prompting guidelines. In *International Conference on Artificial Intelligence in Education*. Springer, 364–377.

[36] Kim Leighton, Suzan Kardong-Edgren, Tonya Schneidereith, and Colette Foisy-Doll. 2021. Using social media and snowball sampling as an alternative recruitment strategy for research. *Clinical simulation in nursing* 55 (2021), 37–42.

[37] Guang Li, Mohammad Amin Zarei, Goudarz Alibakhshi, and Akram Labbafi. 2024. Teachers and educators' experiences and perceptions of artificial-powered interventions for autism groups. *BMC psychology* 12, 1 (2024), 199.

[38] Xiaojuan Li, Xinying Xu, and Jingwen Zhu. 2024. Research on the Application of Multimedia Technology in Classroom Teaching in Special Education Schools. In *IoT and Big Data Technologies for Health Care*. Springer, 91–101.

[39] Yini Liao, Karola Dillenburger, and Xiaoyi Hu. 2022. Behavior analytic interventions for children with autism: Policy and practice in the United Kingdom and China. *Autism* 26, 1 (2022), 101–120.

[40] Yini Liao, E Zou, Ziyan Chen, Sanmei Cheng, Yunqi Zhou, and Lingjie Liu. 2024. Group social skill training for children on the autism spectrum: an exploratory study of a short-term program in a Chinese outpatient clinic. *International Journal of Developmental Disabilities* (2024), 1–17.

[41] Qing Liu, Wu-Ying Hsieh, and Gaowei Chen. 2020. A systematic review and meta-analysis of parent-mediated intervention for children and adolescents with autism spectrum disorder in mainland China, Hong Kong, and Taiwan. *Autism* 24, 8 (2020), 1960–1979.

[42] Yingna Liu, Jialing Li, Qiaolan Zheng, Charles M Zaroff, Brian J Hall, Xiuhong Li, and Yuantao Hao. 2016. Knowledge, attitudes, and perceptions of autism spectrum disorder in a stratified sampling of preschool teachers in China. *BMC psychiatry* 16, 1 (2016), 142.

[43] Chad R Lochmiller. 2021. Conducting thematic analysis with qualitative data. *The qualitative report* 26, 6 (2021), 2029–2044.

[44] Helen McCabe. 2003. The beginnings of inclusion in the People's Republic of China. *Research and Practice for Persons with Severe Disabilities* 28, 1 (2003), 16–22.

[45] Meizi. 2025. When Special Education Teachers Meet AI: From "Confusion" to "Aha Moment". https://mp.weixin.qq.com/s/kFIaTndqEqys6EPFCcf7tg. WeChat public article, in Chinese.

[46] Denis Newman-Griffis, Jessica Sage Rauchberg, Rahaf Alharbi, Louise Hickman, and Harry Hochheiser. 2022. Definition drives design: Disability models and mechanisms of bias in AI technologies. *arXiv preprint arXiv:2206.08287* (2022).

[47] Samuel L Odom, Lana Collet-Klingenberg, Sally J Rogers, and Deborah D Hatton. 2010. Evidence-based practices in interventions for children and youth with autism spectrum disorders. *Preventing school failure: Alternative education for children and youth* 54, 4 (2010), 275–282.

[48] Mieczysław L Owoc, Agnieszka Sawicka, and Paweł Weichbroth. 2019. Artificial intelligence technologies in education: benefits, challenges and strategies of implementation. In *IFIP international workshop on artificial intelligence for knowledge management*. Springer, 37–58.

[49] Classroom Discourse Patterns. 2024. Using AI Tools to Provide Teachers with Fully. *Uses of Artificial Intelligence in STEM Education* (2024), 371.

[50] Katelyn Pryor, Thorleif Coleman, Syed Zuhair Hossain, Eric Tootil, and Shameem Ahmed. 2025. Examining Large Language Models Within Autism-Related Contexts: A Systematic Review of Bias and (Mis) Representation. In *2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC)*. IEEE, 876–886.

[51] Salih Rakap and Serife Balikci. 2024. Enhancing IEP goal development for preschoolers with autism: A preliminary study on ChatGPT integration. *Journal of autism and developmental disorders* (2024), 1–6.

[52] Sivan Schwartz, Avi Yaeli, and Segev Shlomov. 2023. Enhancing trust in LLM-based AI automation agents: New considerations and future challenges. *arXiv preprint arXiv:2308.05391* (2023).

[53] Olivia Ann Sullivan and Chongying Wang. 2020. Autism spectrum disorder interventions in mainland China: A systematic review. *Review journal of autism and developmental disorders* 7, 3 (2020), 263–277.

[54] Xiang Sun and Carrie Allison. 2010. A review of the prevalence of autism spectrum disorder in Asia. *Research in Autism Spectrum Disorders* 4, 2 (2010), 156–167.

[55] Evdokia Voultsiou and Lefteris Moussiades. 2025. A systematic review of AI, VR, and LLM applications in special education: Opportunities, challenges, and future directions. *Education and Information Technologies* (2025), 1–41.

[56] Evdokia Voultsiou, Eleni Vrochidou, Lefteris Moussiades, and George A Papakostas. 2025. The potential of Large Language Models for social robots in special education. *Progress in Artificial Intelligence* (2025), 1–25.

[57] She Wanbin and Wu Yan. 2025. The use of ICT in teaching by special education teachers in Sichuan, China: determinants and solutions to challenges. *Humanities and Social Sciences Communications* 12, 1 (2025), 1–8.
[58] Shen Wang, Tianlong Xu, Hang Li, Chaoli Zhang, Joleen Liang, Jiliang Tang, Philip S Yu, and Qingsong Wen. 2024. Large language models for education: A survey and outlook. *arXiv preprint arXiv:2403.18105* (2024).
[59] Yan Wang, Wangqian Fu, and Sisi Xu. 2025. Special education teachers' emotional labor strategies and its relationship with job burnout in China: The moderating role of psychological capital. *WORK* 81, 2 (2025), 2582–2592.
[60] Yan Wang, Zhiqiang Wang, Nan Zhu, Yajing Feng, Junyi Yin, and Liting Zhao. 2012. An Investigation of the Staffing and Demands of Special Schools in China. *Chinese Journal of Special Education* 11 (2012), 3–8.
[61] Melissa Warr, Nicole Jakubczyk Oster, and Roger Isaac. 2024. Implicit bias in large language models: Experimental proof and implications for education. *Journal of research on technology in education* (2024), 1–24.
[62] Iain Weissburg, Sathvika Anand, Sharon Levy, and Haewon Jeong. 2024. Llms are biased teachers: Evaluating llm bias in personalized education. *arXiv preprint arXiv:2410.14012* (2024).
[63] Tung-Ju Wu, Lian-Yi Wang, Jia-Ying Gao, and An-Pin Wei. 2020. Social support and well-being of Chinese special education teachers—An emotional labor perspective. *International Journal of Environmental Research and Public Health* 17, 18 (2020), 6884.
[64] Xinbo Wu, Junwen Liang, Yansha Dong, Qiuxian Ou, Jiaxi Chen, Linghui Zou, Weixin Dong, and Chunxia Lu. 2025. Effects of VR-Based Serious Games on Gross Motor Skills in Chinese Children with Autism Spectrum Disorder in Special Education: A Pilot Study. *Journal of Autism and Developmental Disorders* (2025), 1–9.
[65] Xinhua. 2021. China sees marked increase in students receiving special education. https://en.people.cn/n3/2021/1003/c90000-9903714.html. People's Daily Online (English), accessed 2021.
[66] Zhiyi Xu. 2024. AI in education: Enhancing learning experiences and student outcomes. *Applied and Computational Engineering* 51, 1 (2024), 104–111.
[67] Hanlin Yang and Hongbo Wang. 1994. Special education in China. *The Journal of Special Education* 28, 1 (1994), 93–105.
[68] Huaiyuan Yao, Wanpeng Xu, Justin Turnau, Nadia Kellam, and Hua Wei. 2025. Instructional Agents: LLM Agents on Automated Course Material Generation for Teaching Faculties. *arXiv preprint arXiv:2508.19611* (2025).
[69] Jinan Zeidan, Eric Fombonne, Julie Scorah, Alaa Ibrahim, Maureen S Durkin, Shekhar Saxena, Afiqah Yusuf, Andy Shih, and Mayada Elsabbagh. 2022. Global prevalence of autism: A systematic review update. *Autism research* 15, 5 (2022), 778–790.
[70] Chunpeng Zhai, Santoso Wibowo, and Lily D Li. 2024. The effects of over-reliance on AI dialogue systems on students' cognitive abilities: a systematic review. *Smart Learning Environments* 11, 1 (2024), 28.
[71] Chong Zhang and Xiaoyan Miao. 2022. Systematic Review on Chinese Special Education and Inclusive Education: China's Solution in Globalization. *ECNU Review of Education* (2022), 20965311221147310.
[72] Hui Zhang, Xueqin Qian, and George Singer. 2022. Experiences of Chinese parents of children with Autism Spectrum Disorder advocating for inclusive education. *International Journal of Special Education* 37, 1 (2022), 62–74.
[73] LING ZHANG, Haidee Jackson, Sohyun Yang, Xueqin Qian, Richard Carter, Jennifer Diliberto, and Jihong Zhang. 2024. Prototyping A Multi-Agent System to Enhance AI-Human Collaboration in Individualized Education Program Development. (2024).
[74] Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, et al. 2025. Siren's Song in the AI Ocean: A Survey on Hallucination in Large Language Models. *Computational Linguistics* (2025), 1–46.
[75] Bingrui Zhou and Xiu Xu. 2019. Progress and challenges in early intervention of autism spectrum disorder in China. *Pediatric Medicine* 2 (2019).
[76] Ce Zhou, Qian Li, Chen Li, Jun Yu, Yixin Liu, Guangjing Wang, Kai Zhang, Cheng Ji, Qiben Yan, Lifang He, et al. 2024. A comprehensive survey on pretrained foundation models: A history from bert to chatgpt. *International Journal of Machine Learning and Cybernetics* (2024), 1–65.
[77] Nan Zhu, Li Gao, Jiaojiao Wang, Yan Wang, Zhonghe Huang, Nannan Guo, and Yajing Feng. 2021. Professional qualities of special education itinerant teachers: A qualitative study from China. *International Journal of Disability, Development and Education* 68, 6 (2021), 742–757.
[78] Qingxiaoyang Zhu and Hao-Chuan Wang. 2023. Leveraging Large Language Model as Support for Human Problem Solving: An Exploration of Its Appropriation and Impact. In *Companion Publication of the 2023 Conference on Computer Supported Cooperative Work and Social Computing*. 333–337.

A ORIGINAL REDNOTE POSTS OF PARTICIPANTS

B INTERVIEW QUESTION LIST

B.1 Overall Experiences and Evaluations of Conversations

- Could you describe your overall experience of interacting with the LLM CAI over the past two weeks? (You may talk about your first impressions, how your feelings evolved during use, and the general atmosphere.) How did you first learn about this kind of tool? Did your perception or feelings toward AI change before and after using it?
- What were your main purposes or goals in using CAI?
- Have you used other similar tools before? What made you choose this AI tool, and do you see any differences?
- Do you think the conversations with AI differ from your conversations with colleagues or other people? What motivates you to chat with AI?
- Looking back on your conversations with it, at which moments did you find it especially helpful, and at which moments did you feel it was stuck or not smooth?

B.2 Perceived Advantages of Interacting with CAI

- One day you mentioned reading a book about using different communication modes flexibly with children. Could you briefly introduce the content of this book and explain why you felt AI could help you with this? When it comes to providing examples, what do you see as the biggest difference between AI's responses and the examples in the book?
- You mentioned that AI feels like a mobile knowledge base. Do you think it is completely accurate, and would you always follow its suggestions?
- You also said AI feels like a peer "listening ear" or a safe space. Why would you choose not to talk about these issues with other colleagues? And why do you feel that AI helps you relieve emotions?
- Why do you think AI is able to help you break out of habitual thinking patterns?
- If you had to choose the single most valuable quality of AI, how would you describe it?

B.3 Challenges Encountered During Use

- During your use of the tool, were there any moments when you felt confused, restricted, or inconvenienced? Could you describe a specific example?
- You mentioned that "when I first started using AI, I also made some mistakes" — what exactly did you mean by this? You also said its responses were "too templated." In what ways did you notice this templated pattern? Why do you think this happened, and what changes made you feel that AI started to become more useful?
- You said that AI cannot directly solve teaching problems. Why do you feel this way? And why do you think AI cannot replace the real experiences of peers?

(A) Original post of P1 **(B) Original post of P5**

Figure 3: Original versions of participant RedNote posts. The translated version is located in Figure 2 of the main text.

- How would you compare AI's companionship with face-to-face companionship? In what ways do you think AI cannot provide the same kind of support?
- In what aspects has AI not met your expectations? What do you wish it could do, but currently cannot?
- Do these challenges affect your willingness to keep using it?

B.4 Ethical Concerns of Interacting with CAI

- During these two weeks, did you ever consider or worry about any ethical issues? Could you describe in which situations you thought about them, and how you dealt with those concerns?
- Do you think avoiding ethical issues is important when using such a tool? Why do you think it matters?
- Were there any moments when the AI's responses conflicted with your own views or the values you hold?
- Did you notice any signs of bias, inappropriate expressions, or lack of sensitivity to the special education context in its responses? Could you give an example?
- If this tool were to be used in schools on a long-term basis, what ethical, privacy, or safety issues do you think would need special attention?

C SUPPLEMENTARY TABLE

Table 3: Diverse Goals and Contexts of SETs' LLM CAI Use (Full)

Goal	Context	Example Prompts	Frequency[1]
Interpreting behaviors and understanding students' intentions	Slow response times (P1); Preference for sticky textures (P7); Behaviors such as touching teachers' arms or peers' faces with lips (P4); Behaviors such as running after peers (P8); Assessing the urgency and level of support required in response to behaviors (P7); Building relationships and social motivation (P2); Supporting emotion regulation (P1, P2, P4, P5, P7)	**[P7 (to ChatGPT)]:** I have a third-grade student with autism. Her tactile seeking behavior involves a preference for sticky things, and her other sensory-seeking behaviors include enjoying pleasant smells, lightly touching others with her lips, running around quickly, liking her hands to be held tightly, and enjoying jumping from heights. What might be causing this?	46
Designing, simulating, validating, and optimizing classroom intervention strategies	Creating social story texts and images (P4, P6, P8); Designing reinforcement tools (P2, P4); Developing picture books to teach daily routines (e.g., washing face) (P6); Recommending appropriate animations to foster moral reasoning (P5, P7); Revising intervention plans for children with slower responses (P1); Integrating children's interests into intervention contexts, goals, and processes (P3, P8); Addressing learning regression (P3); Providing personalized support (in group teaching or home visits) (P9); Addressing challenges faced by autistic children entering puberty (P6, P12); Academic skills (addition and subtraction, comparison games, improving handwriting habits) (P4); Methods for teaching daily living skills to children with poor fine motor ability (P6); Speech therapy (phonics training) (P6, P10, P11); Supporting children with attention difficulties (P2, P3)	**[P3 (to DeepSeek)]:** I currently have an autistic student in my class for whom I need to carry out interventions. I have noticed that he is particularly interested in diamond painting activities. However, I feel that this activity is somewhat monotonous. How can I adjust it?	73
Facilitating communication with other stakeholders	Addressing parents' anxiety of children's progress (P1, P7, P8); Responding to parents' doubts about intervention plans (P1); Providing home-based intervention suggestions (P1, P5); Handling conflicts in perspectives with parents (P1, P5); Ensuring children complete tasks at home (P3); Presenting children's classroom performances to parents (P11); Understanding parents' true concerns or intentions (P4); Helping other teachers understand children's needs (P1, P5, P6, P10); Offering emotional reassurance to colleagues (P2); Navigating collaboration with general education teachers (P8)	**[P1 (to ChatGPT)]:** A few days ago, a child with autism in my class hit a teacher and even said, "I'm going to kill you," which understandably scared the teacher. I think this student's current behavior is partly related to his grandmother, because whenever he acts out, she tends to give in to him. However, it's often very difficult to change the habits of older adults.	54
Getting emotional support	Coping with children's refusal to cooperate or fluctuating progress (P3, P5); Self-doubt triggered by minor incidents or intervention setbacks (P2, P4, P5); Self-emotional regulation and reflection (P2, P3,P10); Sharing personal growth experiences and gaining recognition (P2); Frustration due to lack of parental recognition (P3, P5)	**[P5 (to DeepSeek)]:** Today I conducted an open class, and I felt that my abilities might still be limited. I have been reflecting a lot and also feel quite lost.	14
Seeking personal growth strategies	Career planning for the next three years (P4); Understanding salary progression for shadow teachers with different years of experience (P9); Exploring the development of shadow teachers across countries (P9); Learning about other special education intervention programs (P8)	**[P4 (to Doubao)]:** With the decline in birth rates, what is the future for special education teachers? Please create a three-year career plan for a special education teacher.	5

[1] "Frequency" refers to the total number of conversations across all participants, with each conversation counted once per instance. It was aggregated by summing up all the relevant chat instances related to specific professional goals over the two-week period. This gives an overall count of how many times the LLM-based CAI was used for these professional purposes across the entire group of participants.